

Arslan Khalid

Python Developer

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EDUCATION

B.Sc. Electrical Engineering

University of Gujrat

CGPA: 3.12

Nov 2020 – Jul 2024

Relevant Courses:

- [Advanced ML Algorithms by Stanford University](#)
- [Supervised Machine Learning by Stanford University](#)
- [Django Web Framework by Meta](#)

SKILLS

Languages:

Python · C++

Libraries/Frameworks:

BeautifulSoup · FastAPI · Django · TensorFlow · Hugging Face

Machine Learning:

NLP · Computer Vision · Reinforcement Learning (RL) · RAG Models

Version Control:

Git · GitHub · GitLab

Development:

Clean Code · Deploying ML Models · Python for Automation

Soft Skills:

Team Player · Problem Solving · Critical Thinking · Communication Skills

EXPERIENCE

• GulzarSoft (On-site)

Gujrat, Pakistan

Python Developer

Oct 2024 – Present

- **Car Recommendation System:** Designed and implemented a collaborative filtering model for a car recommendation system, enhancing recommendation accuracy by 30%.

• Coders Launchpad (Remote)

Dubai, UAE

Machine Learning Intern

Nov 2023 – Feb 2024

- **Voice-to-Voice Conversion Model:** Assisted in training a voice-to-voice conversion model for the Quran Project using RVC, contributing to high-quality audio synthesis.
- **Audio Synthesis:** Enhanced audio quality for Quran recitations, improving accuracy and naturalness in voice conversion.

• Intelligent Systems Lab (On-site)

Gujrat, Pakistan

Machine Learning Intern

Sep 2023 – Oct 2023

- **Facial Recognition Model:** Developed a facial recognition model using TensorFlow to match ID card images with live photos, improving identification accuracy.
- **Image Matching:** Leveraged machine learning techniques to accurately match facial features between static and live images, streamlining identification.

PROJECTS

• ZARM — Backend & AI Integration System

FastAPI, PostgreSQL, Redis, OpenAI

- Engineered a high-performance, asynchronous backend architecture utilizing FastAPI, PostgreSQL, and Redis to drive scalable consulting workflows.
- Architected secure, multi-tenant database models to ensure strict data isolation and efficient querying across multiple client environments.
- Integrated advanced AI agent layers into the backend infrastructure to automate data synthesis and client reporting, reducing manual processing time by 85%.

• Jarvis — Real-Time AI Voice Assistant

LiveKit, Python, WebSockets, Wake-Word Detection

- Architected a persistent, low-latency AI voice assistant leveraging LiveKit real-time infrastructure.
- Trained and deployed custom wake-word detection models, optimizing streaming connection latency to achieve near-instantaneous conversational responsiveness.

- Bridged continuous audio streaming with backend AI processing pipelines, enabling seamless and natural user interactions.

- **Project Syndra — Social Media Automation Engine**

Python, Node.js, Docker, Web Scraping APIs

- Developed an end-to-end automated news aggregation and publishing engine tailored for social media platforms using Python and Node.js.
- Built a robust data pipeline to autonomously scrape, dynamically rank, and filter visual news content based on custom relevance algorithms.
- Integrated external APIs to schedule and publish high-quality visual content, completely automating the content pipeline and saving 20 hours of manual curation weekly.

- **Cortex — LLM Reasoning Visualization Tool**

React, TypeScript, Python, FastAPI, WebSockets

- Designed and deployed a full-stack software system dedicated to tracking, logging, and visually mapping the internal reasoning chains of Large Language Models (LLMs).
- Built a highly responsive frontend dashboard and robust backend architecture to process and display complex AI inference steps in real-time.
- Enabled developers to effectively debug, audit, and optimize LLM prompts and agent behaviors through comprehensive visual analytics.

[AWARDS](#)

- Achieved **top 5** position in Devathon, a hackathon organized by Devsinc, Pakistan.
- Earned the **Master** title on Kaggle for excellence in data science and machine learning competitions.
- Received funding for my Final Year Project (FYP) through the **IGNITE NIGRI** program.
- Secured **first position** in 9th, 10th, 11th, and 12th grades for academic excellence.
- Won the **typing competition** organized by 10FastFingers.